DHCP Security

Project Overview

This project was initiated to assess and enhance the security posture of the DHCP services within the network environment. The primary objective is to protect the network from DHCP-based attacks and ensure reliable IP address management. The project focuses on the following goals:

- Evaluate DHCP security risks such as rogue servers, DHCP starvation, and spoofing attacks.
- Implement best practices and security features like DHCP snooping, DHCPv6 guard, and DHCP relay agent policies.
- Define trusted and untrusted ports and VLANs to prevent unauthorized DHCP traffic.
- Establish monitoring and logging procedures for DHCP activities to detect anomalies early.
- Provide clear guidelines and policies to maintain DHCP security and operational continuity.

The project outcome delivers a robust and secure DHCP infrastructure that supports stable network connectivity and mitigates common DHCP-related threats.

Description

Dynamic Host Configuration Protocol (DHCP) operates at the application layer and is used to dynamically assign Internet Protocol (IP) addresses to each host on your organization's network. Because DHCP runs over UDP and because one side of all UDP communication does not have an IP address during the conversation, DHCP is an inherently insecure protocol. While current DHCP implementations have many vulnerabilities, there are many current proposals for improving the security of the protocol.

Recommendations

To mitigate the risks associated with DHCP, the assessment team recommends the following steps:

- Enable DHCP Snooping to prevent rogue DHCP servers and starvation attacks.
- Configure Port Security and rate limiting to block excessive IP requests.
- Avoid placing DHCP on domain controllers to reduce attack surface.
- Implement DHCP Failover for high availability and reliability.
- Minimize use of static IPs to simplify network management.
- Consider Authentication Protocols
- Use a Non-Domain Controller Member Server for DHCP

References

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Details

Most common DHCP attacks are:

- DHCP Starvation – Attacker floods DHCP server with requests to exhaust IP pool.
- Rogue DHCP Server – Attacker sets up fake DHCP server to hand out bad configurations.
- DHCP Spoofing – Rogue server assigns attacker as default gateway to intercept traffic.
- DHCP DoS Attack – Prevents clients from receiving valid IP configs (denial of service).
- Misconfiguration Exploit – Exploits weak or misconfigured DHCP settings to redirect traffic.

To increase network security, network administrators may adopt a multilayered approach when protecting Dynamic Host Configuration Protocol systems from threats.

- Authentication and access control for DHCP servers and clients stops rogue DHCP servers and ensures that only authorized clients can receive IP addresses.
- Firewalls can monitor and filter traffic, and secure the DHCP server from unauthorized access or attacks.
- Logging enables administrators to monitor the performance of DHCP servers and identify suspicious behavior or anomalies.
- Applying patches and updating servers can help prevent attacks that exploit vulnerabilities.
- Data encryption mitigates data breaches and prevents eavesdropping.
- DHCP snooping filters out rogue DHCP messages.
- DNS firewalls block malicious domains or IP addresses.

DHCP Snooping

DHCP snooping protects the network from common DHCP attacks, including address spoofing resulting from a rogue DHCP server operating on the network or exhaustion of addresses on a DHCP server caused by mass address requests by an attacker on the network. DHCP snooping designates trusted DHCP servers and ports on which DHCP requests and responses are accepted.

DHCPv6 Guard:

DHCPv6 Guard enhances DHCPv6 Snooping by creating policies with specific rules applied on VLANs and ports. Packets on trusted ports are checked against these policies; compliant packets are forwarded, others dropped.

DHCP Snooping Binding Database:

DHCP snooping builds and maintains a database (binding table) of untrusted hosts with leased IPs on VLANs where snooping is enabled. Trusted ports are excluded. Entries are added when DHCPACK messages are received and removed when leases expire or DHCPRELEASE is received. Each entry includes MAC address, IP, lease time, VLAN, and interface info. This database is also used by Dynamic ARP Inspection and IP Source Guard. Entries can be cleared with clear ip dhcp snooping binding.

DHCP Relay Agent:

You can configure the device to run a DHCP relay agent, which forwards DHCP packets between clients and servers. This feature is useful when clients and servers are not on the same physical subnet. Relay agent forwarding is distinct from the normal forwarding of an IP router, where IP datagrams are switched between networks somewhat transparently. By contrast, relay agents receive DHCP messages and then generate a new DHCP message to send out on another interface.

To configure basic DHCP snooping:

1. Log into the switch and enter configuration mode (conf t).
2. Enable DHCP snooping with ip dhcp snooping.
3. Specify VLAN(s) to monitor using ip dhcp snooping vlan [VLAN_ID].
4. Set trusted ports by entering the port config (int g0/24) and using ip dhcp snooping trust.

Affected Locations

- Client devices
- Switch ports
- DHCP servers
- Network VLANs
- Router interfaces
- Network segments/subnets